QuickPlot

SWE 4404: Software Project Lab II

Group 09

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Requirement Identification



Elicitation

- Domain study of Graphing tools
- Info from Students, Educators, Researchers



Analysis

- **SoC:** Parser → Evaluator → Renderer
- Prioritization



Specification

- FRs refined into User stories
- Acceptance Criteria via Google Forms



Validation

- Peer and Supervisor Review
- Feasibility check according to the plan

Separation of Concerns · Abstraction · Decomposition · Modularity

Actors & Use Cases

Visitors(Guest)

Unregistered; can graph freely and register

Registered User

Has an account; everything a visitor can, with persistence

Link Viewer

Opening a shared URL; views the exact workspace without logging in

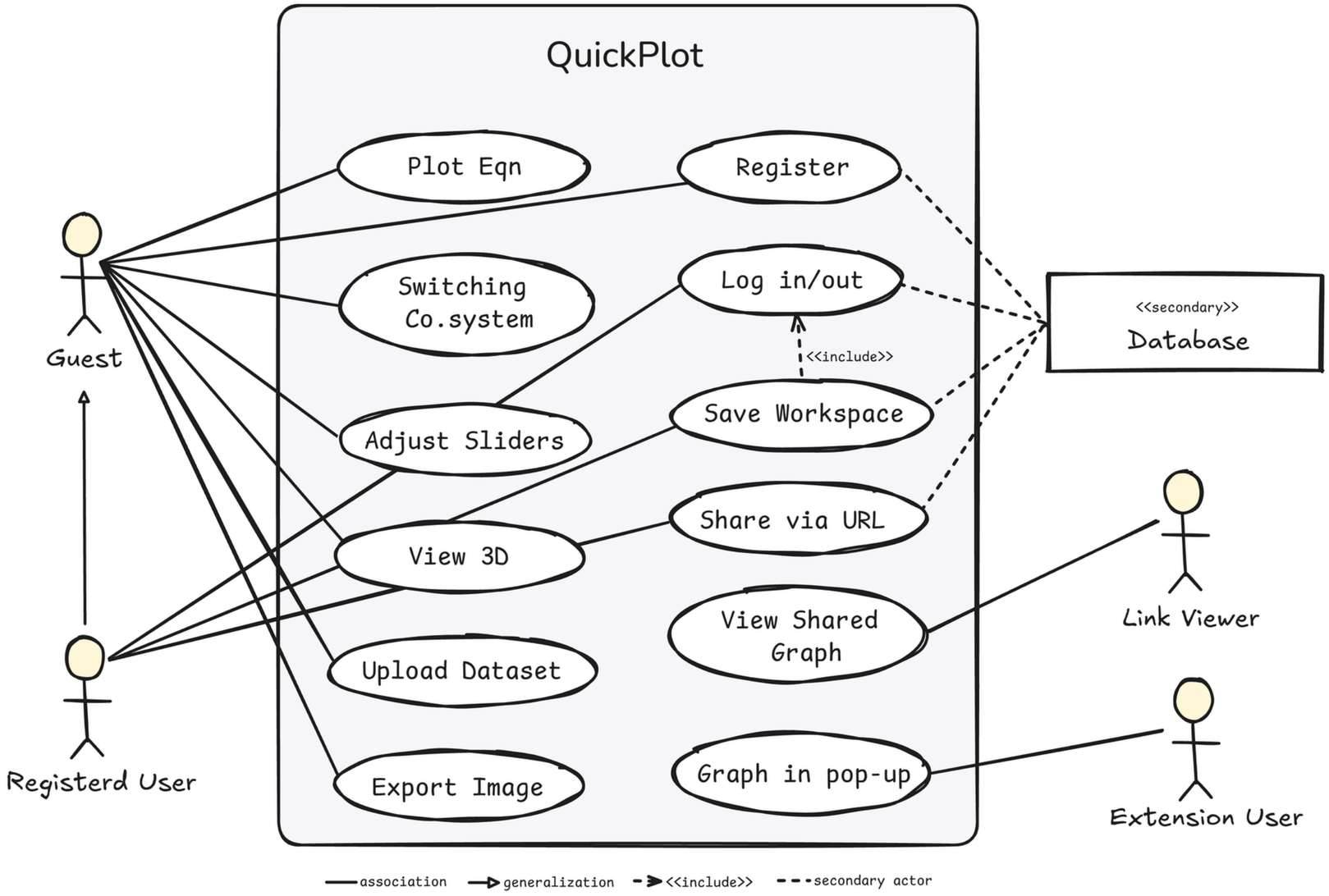
Extension User

Graphs from any browser tab pop-up

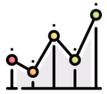
Database

Stores workspaces, Accounts; drawn on the right per UML convention

Actors & Use Cases: Diagram



Functional Requirements



Real-Time Grapher

- Accepts Standard Notations
- Updates within 500ms
- Error message on invalid input



Coordinate Systems

- Switch between co-ordinates
- Updating Axis level



Parameter Sliders

- Auto-detect the Parameters
- Slider is editable
- Simulation Button



3D Surface

- Toggle to 3D canvas
- Orbit control for Rotate & Zoom
- Fallback text on Technical issues



Dataset Upload

- Multiple Datasets
- Wrong Data detection



Browser Extension

- Plot via copying
- Graph on pop-up UI

Requirement Prioritization (MoSCoW)

Must Have

- Real-time grapher
- Cartesian ↔ Polar
- Parameter sliders
- 3D surface plots

Should Have

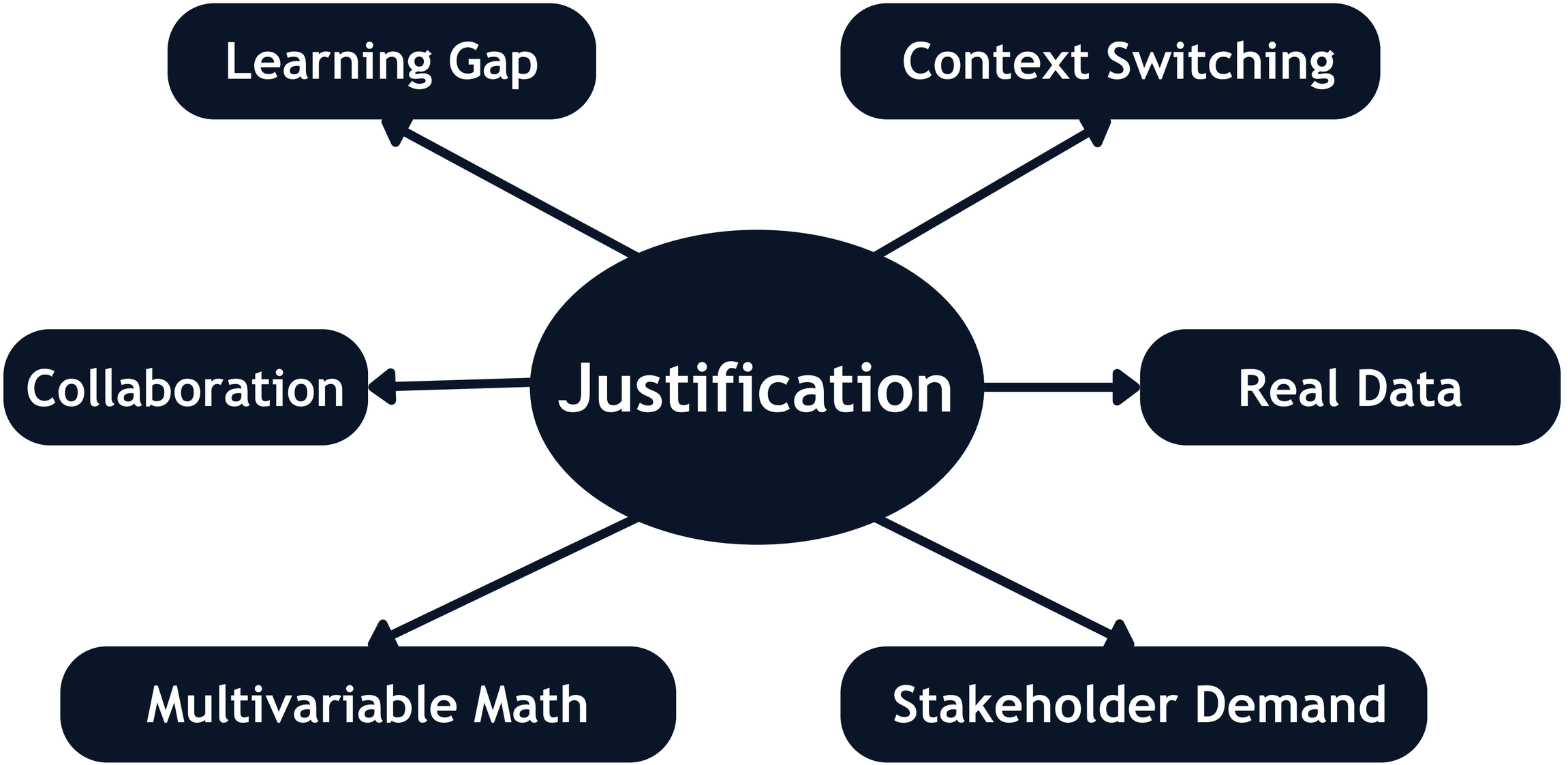
- Register Accounts
- Save workspace
- Share via URL
- Export PNG
- Dataset upload

Could Have

- Canvas themes
- Browser extension

Won’t Have

- Mobile app
- Real-time collaboration
- Offline mode


Learning Gap
Context Switching
Collaboration
Justification
Real Data
Multivariable Math
Stakeholder Demand

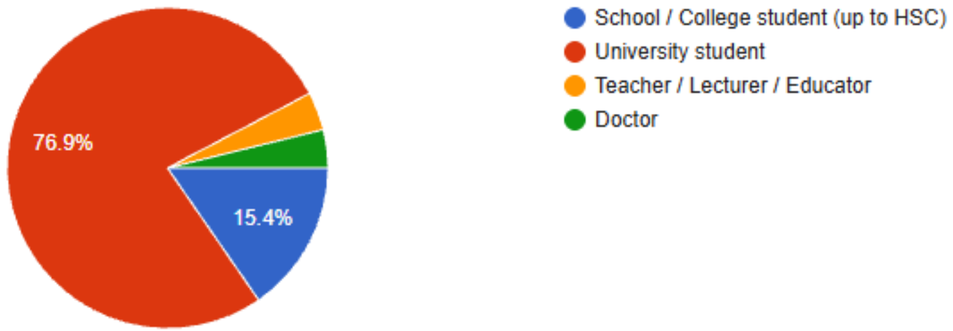
Thank You

Any Feedbacks?

Referrence

Google forms

https://forms.gle/E8Q514ud55cUfCBx8



SRS Spreadsheet